

Reference Textbooks

- Computer Security: Principles and Practice (SPP), 4th ed. by William Stallings and Lawrie Brown
- Cryptography and Network Security (CNS), 8th ed. by William Stallings
- Principles of Information Security, 6th ed. by M. Whitman and H. Mattord
- CISSP All-in-one exam Guide, 9th ed. by Maymi, Harris
- Computer Security Fundamentals, 5th ed. by Chuck Easttom

Tentative Grading Criteria

- Assignments 8%
- Quizzes 10%
- Midterm Exams 30%
- Project 12%
- Final Exam 40%

Grading Scheme: **Absolute**

Tentative Weekly Schedule

Lec #	Content Covered
1	Course Introduction • Introducing syllabus, policies, and projects. • An overview of information security goals: confidentiality, integrity, availability, authentication, authorization and non-repudiation. • Common attacks against security
2	Security Design Principles Discussion and evaluation of following primitives: Least-privilege, fail-safe defaults, complete mediation, separation of privilege, economy of mechanism, open design, psychological acceptability, defense in depth, compartmentalization
3	Classical Cryptography Introduction to Cryptography: Symmetric cipher model Classical substitution techniques (Caesar, Monoalphabetic cipher)
4	Classical Cryptography II Classical substitution techniques (Vigenère cipher, One-time pad) Transposition techniques (Rail fence cipher, Columnar transposition cipher)
5	Modern Ciphers Block cipher structure and design principle Stream ciphers introduction
6	Modern Ciphers II AES structure and transformations Block cipher modes of operation (ECB, CBC, CTR)

7	Asymmetric Cryptography Introduction to Public Key cryptography RSA principles and algorithm Hybrid cryptosystem: digital envelope
8	Asymmetric Cryptography II Diffie-Hellman key exchange algorithm Man-in-the-middle attack against Diffie-Hellman
9	Message Integrity and Authentication Message Authentication Code (MAC) schemes HMAC algorithm & structure Cryptographic hash func. properties (preimage resistance, collision resistance) Digital signatures workflow
10	Public Key Infrastructure (PKI) Public key certificates (X.509) Elements of PKI Chain of trust Certificate revocation
First Mid-term Exam	
11	Malicious Software I Malware, types (virus, worms, trojans, adware, spyware, backdoor, ransomware, rootkits, bootkits) Virus classification based on payloads, targets and obfuscation
12	Malicious Software II Rootkit and bootkit operation Static and dynamic analysis of malware samples
13	Memory Safety Vulnerabilities Buffer overflow attacks Compile time and run time defenses
14	Database Security: SQLi SQL Injection attack: techniques, types SQLi Countermeasures
15	Web Security: XSS Background of web security Cross Site Scripting (XSS) Attack Types of XSS (reflected, stored, DOM based) Countermeasures of XSS
16	LLM Security Prompt Injection attacks Injection types: Direct, Indirect with examples Injection impacts: prompt leaks, data theft, misinformation spread etc. Defense and Mitigation

17	User Authentication Knowledge based authentication (passwords) Password cracking attacks Salted password storage Challenge-response authentication
18	User Authentication II Possession based authentication (OTPs) Biometric authentication Multi-service authentication: Kerberos overview & key exchange protocol
19	Access Control Access control policies Discretionary, Mandatory, Role-based Access Control Example: GRANT and REVOKE statements in SQL Attribute based access control
20	Network Security Transport Layer Security (SSL/TLS) TLS handshake, record protocols
Second Mid-term Exam	
21	Network Security II Internet Protocol Security (IPSec), IPsec modes (transport, tunnel) Authentication Header (AH), Encapsulated Security Payload (ESP)
22	Intrusion Detection Intrusion Detection Systems (IDS) components Classification of IDS (anomaly, signature, hybrid) Types (host-based, network based) Deception & honeypots
23	Network Security: Firewalls Firewalls Types (packet-filtering, stateful, circuit-level proxy, DPI) Placement of firewall UTM firewalls
24	Security Operations Security Operations Center (SOC) Roles of SOC Analyst and SOC Engineer Security Information and Event Management (SIEM) systems Threat Intelligence Feeds
25	IT Security Management Security Policy documents Risk management framework Contingency planning: Incident response, Disaster recovery

26	Cybercrime Laws and Ethics Pakistan cybercrime act and the role of investigative agencies International laws Ethical issues in IT security Situation analysis using ethical principles
27 - onwards	Revision & Project Evaluations